



PATHOLOGY
PATHOLOGY
pathology
病理學

學號：

Vol.

姓名：

BLOCK 1 考古題本

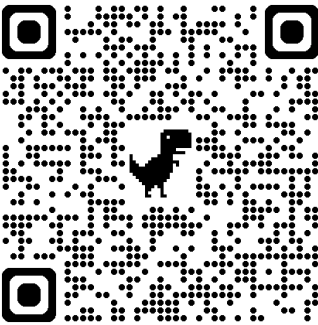
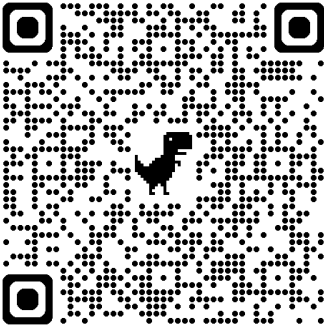
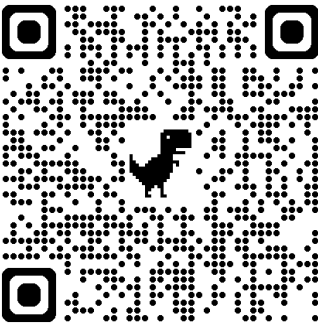
共筆製作群		
寫手： 黃律勳、林庭寬、李彥熙、 蓮怡安、莊棋淳、賈子謙、 丁俊呈、林紓宇、戴源成	組長： 李承洋	共筆長： 郭祐維 王泓宇 康元輔 蕭婷云

重要公布事項：

寫手想跟你 / 妳各位說：

組長想跟你 / 妳各位說：

共筆長想跟你 / 妳各位說：

勘誤表單：	勘誤表	想說的話
		

--

目錄

舊考古	1
§ 鄭永銘老師—緒論	1
§ 吳木榮老師—病理解剖	2
§ 袁章祖老師—細胞病理（一）（二）（三）（四）	6
§ 謝明書老師—急性與慢性發炎（一）（二）	17
§ 謝明書老師—組織修復（一）（二）	23
§ 連晃駿老師—血行動力學病理（一）（二）（三）	30
B05 第 1 次區段考.....	39
B06 第 1 次區段考.....	45
B07 第 1 次區段考.....	50
B08 第 1 次區段考.....	57
B09 第 1 次區段考.....	61
B10 第 1 次區段考.....	67

舊考古

§ 鄭永銘老師—緒論

1. Which of the following is not included in a disease process?
 - (A) Etiology
 - (B) Pathogenesis
 - (C) Morphologic changes
 - (D) Clinical significance
 - (E) Management
2. Which of the following is not part of practice anatomical pathology?
 - (A) Cytopathology
 - (B) Medical genetics
 - (C) Forensic pathology
 - (D) Surgical pathology
3. 配合題：
 - (1) General pathology
 - (2) Special / Systemic pathology
 - (3) Surgical pathology
 - (4) Pathogenesis
 - (A) Mechanism of disease development
 - (B) Integration of clinical findings & pathologic anatomy
 - (C) Examines of the specific responses of specialized organs to well-defined stimuli
 - (D) Concern with basic reactions of cells to abnormal stimuli that underline all disease

§ 吳木榮老師—病理解剖

1. At autopsy, the instrument, “hammer with hook”, is designed for the removal of:
 - (A) Ramus of mandible
 - (B) Patella
 - (C) Calvarium
 - (D) Spinal cord
2. The term, virtopsy, means:
 - (A) Radiographic autopsy
 - (B) Microscopic autopsy
 - (C) Pharmacological autopsy
 - (D) Surgical autopsy
3. Which one is the purpose of autopsy?
 - (A) Doctor's wish
 - (B) Experimental human trial
 - (C) Improve patient's life quality
 - (D) To avoid repeating errors in diagnosis and treatment
4. A surgical pathologist requires the following except:
 - (A) Must know his own field thoroughly and also must have a rich background in clinical medicine
 - (B) Need to understand clinician's need and respond to them
 - (C) Needs for subspecialization
 - (D) Should not communicate with clinicians constantly
5. Which one has the highest risk when performing an autopsy?
 - (A) A patient with myocardial infarction
 - (B) A patient with appendicitis
 - (C) A patient with neurofibromatosis
 - (D) A patient with SARS
6. The performance of virtopsy needs:
 - (A) Computed tomography scanner
 - (B) Microscope
 - (C) Autopsy room
 - (D) Operative room

7. The hospital autopsy needs:
 - (A) Physician's consent
 - (B) Treatment drugs
 - (C) Professor at large
 - (D) Personal or families' consent
8. Which one is an autopsy?
 - (A) Dissection of organs of body
 - (B) Pathological examination of heart and lung
 - (C) Laboratory investigation of body fluids
 - (D) Full pathological examination of organs of body
9. Why do medical students need autopsy?
 - (A) To explore the cause of death
 - (B) To improve operative skills
 - (C) For teaching and learning medical knowledge and experience
 - (D) For professional research
10. Which disease can be transmitted from the dead body during autopsy?
 - (A) Ebola hemorrhagic fever
 - (B) Kaposi's sarcoma
 - (C) Aplastic anemia
 - (D) Squamous cell carcinoma
11. The fall of autopsy is mainly due to:
 - (A) No money to offer
 - (B) No technician for help to perform autopsy
 - (C) No interest at all in physicians
 - (D) No research reward to families
12. Pathological autopsy must have a consent of:
 - (A) Physician
 - (B) Medical administrator
 - (C) Next of kin
 - (D) Social worker
13. The difference between autopsy and dissection is:
 - (A) The site of cutting during both procedures
 - (B) The purpose of both procedures
 - (C) The instrument of both procedures
 - (D) The protection of both procedures

14. Which one is the best choice of the candidate for pathological autopsy in the hospital?
 - (A) Patient with chronic renal failure
 - (B) Patient with hypertension
 - (C) Patient with a gastric ulcer
 - (D) Unknown cause of sudden death during treatment
15. Which disease could be transmitted during autopsy?
 - (A) Hepatitis B virus infection
 - (B) Pneumonia
 - (C) Pemphigus
 - (D) Cerebellar atrophy
16. At autopsy, the time of hospital autopsy by law is:
 - (A) Any time as you wish
 - (B) Within a half of hour
 - (C) Later than 6 hours
 - (D) 2-3 hours in between
17. According to the medical record, the first autopsy report in Taiwan was a patient died of?
 - (A) Aortic aneurysm
 - (B) Hepatic cirrhosis
 - (C) Syphilis
 - (D) Alcohol intoxication
18. Which one is an infectious disease during lung biopsy?
 - (A) Asthma
 - (B) Pneumonia
 - (C) Tuberculosis
 - (D) COPD
19. Which one is the major purpose of autopsy?
 - (A) Organ preservation
 - (B) Medical teaching
 - (C) Doctor's interesting
 - (D) Tissue bank
20. The final pathological of full body postmortem examination is:
 - (A) DNA genomic testing
 - (B) Autopsy
 - (C) 3D reconstruction of gross surgical specimen
 - (D) Digital pathological examination

21. A board-certified physician can perform autopsy at his clinical bath tub?
 - (A) Yes, he can and is well-done.
 - (B) Yes, he can and needs a pathologist's assistance.
 - (C) He cannot and it is illegal.
 - (D) No, he cannot except to report to the in-charge district prosecutor.
22. "See for oneself" is a described method of?
 - (A) Autopsy
 - (B) Medical ethic
 - (C) Medical esthetics
 - (D) Medical esthesiography
23. What is the major cause of the decline of the autopsy rate?
 - (A) Religionism
 - (B) Poor autopsy technique
 - (C) Little research value based on autopsy
 - (D) No interest in physicians and pathologists

§ 袁章祖老師—細胞病理（一）（二）（三）（四）

1. （多選） Which of the following nuclear changes can be seen in necrotic cells?
 - (A) Karyorrhexis
 - (B) Karyolysis
 - (C) Chromatin condensation
 - (D) Pyknosis
 - (E) DNA fragmentation
2. （多選） Which of the following cellular responses observed by light microscopy indicate reversible cell injury?
 - (A) Fatty change
 - (B) Cellular swelling
 - (C) Pyknosis
 - (D) Apoptosis
 - (E) Mitochondrial swelling
3. （多選） The adaptive change expected in the endothelium mucosa in a woman who takes unopposed estrogen therapy is the same type of growth alteration you would expect in which of the following clinical-pathological conditions?
 - (A) Kidney in a patient with end stage renal disease
 - (B) Bronchial epithelium in a patient with smoking history
 - (C) Brain in a patient with stroke
 - (D) Epidermis in skin warts caused by papillomavirus infection
 - (E) Prostate in an old man with urinary retention
4. （多選） Which of the following statements about receptor-mediated apoptosis and mitochondria-mediated apoptosis are correct?
 - (A) Caspase cascade in receptor-mediated apoptosis begins with procaspase-9.
 - (B) Mitochondria-mediated apoptosis makes use of executioner caspases.
 - (C) FADD serves as a protein phosphatase acting on recruited procaspases in receptor-mediated apoptosis.
 - (D) Bid, a proapoptotic BH3-only protein, can be only activated in mitochondrial pathway.
 - (E) Mitochondria-mediated apoptosis involves the movement of cytochrome C into the cytoplasm.

5. (多選) Hyperplasia
 - (A) results in increased cell size.
 - (B) results in increased cell numbers.
 - (C) occurs in prostate gland.
 - (D) occurs in myocardium.
6. Hypoxia, strictly defined, means:
 - (A) failure of intracellular oxidative phosphorylation for any reason.
 - (B) too little perfusion of the tissues.
 - (C) too little oxygen reaching the mitochondria.
 - (D) too little oxygen in the lungs.
 - (E) too little oxygen obtained by an organism.
7. One difference between receptor-mediated apoptosis and mitochondria-mediated apoptosis is that
 - (A) Receptor-mediated apoptosis makes use of procaspase 9.
 - (B) FADD and TRADD serve as protein kinases in receptor-mediated apoptosis.
 - (C) Mitochondria mediated apoptosis make use of executioner caspases.
 - (D) TNF stimulates mitochondria mediated apoptosis.
 - (E) Mitochondria mediated apoptosis involves the movement of cytochrome C into the cytoplasm.
8. Which of following intracellular accumulations is a kind of mutated protein?
 - (A) Hyaline cyst
 - (B) Huntington inclusion body
 - (C) Neurofibrillary tangles
 - (D) Lipofuscin
 - (E) Mallory body
9. In apoptosis the caspases cascade serves to
 - (A) cross-linking the protein.
 - (B) amplify the signal leading to the final cleavage.
 - (C) induce cytochrome c release from mitochondria.
 - (D) mediate crosstalk between intrinsic and extrinsic pathway.
 - (E) bind anti-apoptotic Bcl-2 family proteins.

10. Autophagy 發生的順序：

- (1) Autophagosome fuses with lysosome.
- (2) Cytosolic contents are wrapped with membrane.
- (3) Contents of autophagosome are completely digested.
- (4) Autolysosome membrane ruptures.
- (5) Starvation inhibits mTOR.

- (A) (5)(2)(1)(3)(4)
- (B) (5)(2)(4)(1)(3)
- (C) (5)(2)(1)(4)(3)
- (D) (5)(4)(2)(1)(3)

11. Identify the components of the death-inducing-signaling-complex (DISC).

- (A) NGF, CD95, FADD/TRADD, procaspase-8
- (B) Ligand, CD95, FADD/TRADD, procaspase-8
- (C) Ligand, CD95, FADD/TRADD, procaspase-9
- (D) Ligand, CD95, Apaf-1, procaspase-9
- (E) NGF, CD95, Apaf-1, procaspase-8

12. A pathologist reports fat necrosis after examining a slide. Fat necrosis might be found in which one of the following situations?

- (A) Trauma to the chest wall
- (B) Trauma to the bowel
- (C) Trauma to breast
- (D) Muscle injury
- (E) Brain injury

13. The integral components strongly associated with the intrinsic and extrinsic pathways are respectively:

- (A) The mitochondria and “death receptors”
- (B) “Death receptors” and the mitochondria
- (C) Effector caspases and initiator caspases
- (D) The mitochondria and Bcl-2 family proteins
- (E) The mitochondria and p53

14. Degradation of cells by their own enzymes after an animal has died is called

- (A) Autophagy
- (B) Autodestruct
- (C) Auto genocide
- (D) Autolysis
- (E) Programmed cell death

15. Which of the following statements about telomeres is TRUE?
 - (A) Telomerase activity is shut down in many kinds of human cancers.
 - (B) Telomeres are found in the center of eukaryotic chromosomes.
 - (C) Telomeres do not contain functional genes.
 - (D) Telomeres are degraded by telomerase.
 - (E) All of the above statements about telomeres are false.
16. An autopsy of an 80-year-old woman, dying of old age, revealed that the heart was small (250g) and dark-brown in color on section. Microscopically, there was light-brown perinuclear pigments seen after H&E staining of the cardiac muscle fibers. Which of the following substances is most likely to be increased in the myocardial fibers to produce this cardiac appearance?
 - (A) Calcium, following necrosis.
 - (B) Cholesterol, as a consequence of atherosclerosis.
 - (C) Glycogen, resulting from long-term intracellular storage.
 - (D) Hemosiderin, resulting from iron overload.
 - (E) Lipofuscin, from wear and tear.
17. Some members of the family are pro-apoptotic and promote release of apoptotic factors from the mitochondria, others are anti-apoptotic. From the options below which are the pro-apoptotic family members?
 - (A) Bax and Bcl-2
 - (B) Bid and Bax
 - (C) Bid and Bcl-2
 - (D) Bak and Bcl-xL
 - (E) Bcl-xL and Bid
18. UV radiation is a “negative signal” which can induce apoptosis in cells as a result of various forms of cell damage. If DNA damage is irreparable which of the following is activated as part of the apoptotic cascade?
 - (A) Fas
 - (B) p53
 - (C) c-FLIP
 - (D) Caspase 8
 - (E) All of the above

19. (多選) Which of the following statements are correct?
- (A) Necrotic cells show increased eosinophilia in H&E stains, attributable in part to the loss of cytoplasmic RNA and in part to denature cytoplasmic proteins.
 - (B) Nuclear changes of necrosis appear in one of three patterns: karyolysis, pyknosis, and karyorrhexis.
 - (C) Coagulative necrosis is a form of necrosis in which the architecture of dead tissues is no longer preserved.
 - (D) Ischemia caused by obstruction in a vessel may lead to coagulative necrosis of the supplied tissue in all organs except the brain.
 - (E) Gangrenous necrosis is a specific pattern of cell death denoting to liquefactive necrosis of a limb after injection.
20. (多選) Which of the following statements are correct?
- (A) Caseous necrosis is encountered most often in foci of tuberculous infection.
 - (B) Fibrinoid necrosis is a special form of necrosis usually seen in immune reactions involving blood vessels.
 - (C) Liquefactive necrosis is commonly seen in focal bacterial or, occasionally, fungal infections.
 - (D) Cellular swelling is the first manifestation of almost all forms of injury to cells.
 - (E) The morphologic appearance of necrosis as well as necroptosis is the result of denaturation of intracellular proteins and enzymatic digestion of the lethally injured cell.
21. (多選) Which of the following statements are correct?
- (A) Metaplasia is an irreversible change in which one differentiated cell type (epithelial or mesenchymal) is replaced by another cell type.
 - (B) Metaplasia often represents an adaptive response in which one cell type that is sensitive to a particular stress is replaced by another cell type that is better able to withstand the adverse environment.
 - (C) A deficiency of vitamin A (retinoic acid) induces osseous metaplasia in the respiratory epithelium.
 - (D) In Barrett esophagus, the esophageal squamous epithelium is replaced by intestinal-like columnar cells under the influence of refluxed gastric acid.
 - (E) The influences that predispose to metaplasia, if persistent, can initiate malignant transformation in metaplastic epithelium.

22. (多選) Which of the following statements are correct?

- (A) Necroptosis is triggered by genetically programmed signal transduction events that culminate in cell death.
- (B) Similar to apoptosis, the genetic program that drives necroptosis results in caspase activation.
- (C) Signaling and molecular interactions in necroptosis involves two unique kinases called receptor associated kinase 1 and 3 (RIP1 and RIP3).
- (D) Pyroptosis, via inflammasome, involves activation of caspase-1 which cleaves the precursor form of IL-2 to generate biologically active IL-2.
- (E) Autophagy involves sequestration of cellular organelles into cytoplasmic autophagic vacuoles (autophagosomes) that fuse with lysosomes and digest the enclosed material.

23. (多選) Which of the following statements about apoptosis are correct?

- (A) The unfolded protein response may induce apoptosis.
- (B) In the intrinsic pathway, BCL2 protein is anti-apoptotic, while BAX is proapoptotic.
- (C) Cytochrome c, once released into the cytosol, binds to APAF-1 to form apoptosome, which then activates caspase and results in cell apoptosis.
- (D) The extrinsic pathway is initiated by engagement of plasma membrane death receptors on a variety of cells.
- (E) Chromatin condensation is the most characteristic feature of apoptosis. The chromatin aggregates peripherally, under the nuclear membrane, into dense masses of various shapes and sizes.

24. Which of the following statements is incorrect?

- (A) Metaplasia is a reversible change in which one differentiated cell type (epithelial or mesenchymal) is replaced by another cell type.
- (B) Metaplasia often represents an adaptive response in which one cell type that is sensitive to a particular stress is replaced by another cell type that is better able to withstand the adverse environment.
- (C) In Barrett esophagus, the esophageal squamous epithelium is replaced by another cell type that is better able to withstand the adverse environment.
- (D) The influences that predispose to metaplasia, if persistent, can initiate dysplastic change (or malignant transformation) in metaplastic epithelium.
- (E) All of the above.

25. Which of the following statements about apoptosis is incorrect?

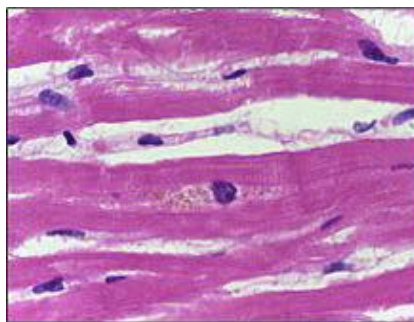
- (A) In the intrinsic pathway, BCL2 protein is anti-apoptotic, while BAX is pro-apoptotic.
- (B) Cytochrome c, once released into the cytosol, binds to BH3-only proteins to form apoptosome, which then activates caspase and results in cell apoptosis.
- (C) The extrinsic pathway is initiated by engagement of plasma membrane death receptors on a variety of cells.
- (D) Chromatin condensation is the most characteristic feature of apoptosis. The chromatin aggregates peripherally, under the nuclear membrane, into dense masses of various shapes and sizes.

26. Which of the following statements is correct?

- (A) The terms steatosis and fatty change describe abnormal accumulations of cholesterol within the interstitium.
- (B) When there is a local or systemic excess of iron, ferritin forms hemosiderin granules, a hemoglobin-derived, golden yellow-to-brown, granular or crystalline pigment.
- (C) Dystrophic calcification is encountered in viable patients of hypercalcemia.
- (D) Metastatic calcification occurs in cancer tissues involving the gastric mucosa, kidneys, lungs, and vessels.

27. 請看右圖

- (A) Hemosiderin
- (B) Melanin
- (C) Ochronosis
- (D) Lipofuscin



28. Which of the following pairings about special stains is incorrect?

- (A) PAS - fungi
- (B) DPAS - glycogen
- (C) Congo red - amyloid
- (D) Trichrome - collagen and muscle

29. Which is not unfolded protein response?

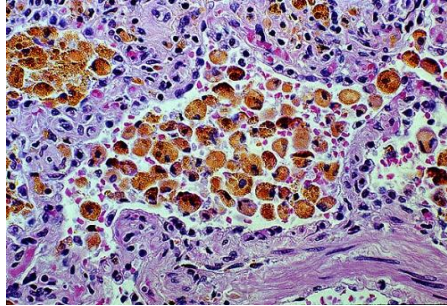
- (A) Increasing the production of chaperones
- (B) Enhancing proteasomal degradation of abnormal proteins
- (C) Triggering microautophagy by lysosome
- (D) Slowing protein translation

30. The causes of cellular aging do not include:

- (A) Accumulation of DNA damage
- (B) Telomere attrition
- (C) Caloric restriction
- (D) Defective protein homeostasis

31. 請看右圖

- (A) Hemosiderin
- (B) Melanin
- (C) Ochronosis
- (D) Lipofuscin



32. If a substrate is PAS (+), DPAS (+) and GMS (+), which of the followings is the most likely answer?

- (A) Glycogen
- (B) Fungi
- (C) Mucin
- (D) Melanin

33. Which kind of adaption is the main mechanism of liver regeneration after partial resection?

- (A) Hypertrophy
- (B) Hyperplasia
- (C) Metaplasia
- (D) Atrophy

34. Which kind of morphological changes are NOT seen in reversible cell injury?

- (A) Myelin figures by EM
- (B) Surface bleb by LM
- (C) Karyorrhexis by LM
- (D) Mitochondrial swelling by EM

35. Which kind of necrotic patterns is most commonly seen in vasculitis?

- (A) Coagulative necrosis
- (B) Liquefactive necrosis
- (C) Caseous necrosis
- (D) Fibrinoid necrosis

36. Which of the following statement is NOT true?
- (A) Calcium efflux from cells is one of the important mechanisms of cellular injury or death.
 - (B) Catalase will degrade hydroxyl peroxides.
 - (C) Release of cytochrome c from mitochondria is one of the important mechanisms of cellular injury or death.
 - (D) Plasma membrane rupture mainly causes necrosis rather than apoptosis.
37. Which of the following ones serves as an anti-apoptosis molecule?
- (A) BID
 - (B) BAX
 - (C) BCL-2
 - (D) BAK
38. What is true about cell death?
- (A) Necrosome is the main mediator of necrosis.
 - (B) Caspase1 is the main mediator of apoptosis.
 - (C) Pyroptosis will not cause inflammation.
 - (D) Necroptosis is initiated by death receptor.
39. Which of the following intracellular or extracellular accumulation is round eosinophilic halaline appearance in HE stain?
- (A) Amyloidosis
 - (B) Russel body
 - (C) Neurofibrillary tangle
 - (D) Mallory-Denk body
40. Which of the following organs is NOT a common site for metastatic calcification?
- (A) Stomach
 - (B) Kidney
 - (C) Lung
 - (D) Pancreas
41. Which of the following is most likely to show saponification?
- (A) Caseous necrosis
 - (B) Pancreatic fat necrosis
 - (C) Gangrenous necrosis
 - (D) Fibrinoid necrosis

42. Which of the following morphological changes is usually not related to cell injury?
 - (A) Surface blebs
 - (B) Cytoplasmic basophilia
 - (C) Myelin figure
 - (D) Vacuolar degeneration
43. Which kind of adaption is the main mechanism for the increase of muscle mass after regular exercise?
 - (A) Hyperplasia
 - (B) Hypertrophy
 - (C) Metaplasia
 - (D) Atrophy
44. 細胞中出現 misfolding protein 時會進行何種細胞凋亡現象？
 - (A) Necrosis
 - (B) Apoptosis
 - (C) Necroptosis
 - (D) Pyroptosis
45. 下列有關 apoptosis 的敘述何者錯誤？
 - (A) Caspase 9 is the initiator of intrinsic pathway.
 - (B) Executioner, including caspase 7, will cause cytoskeleton degradation.
 - (C) BID is a sensor in extrinsic pathway.
 - (D) Cell membrane remains intact in apoptosis.
46. Which of the following molecules is most important for macroautophagy?
 - (A) Necrosome
 - (B) TNFR1
 - (C) LC3
 - (D) Chaperone
47. Which of the following pigments is not yellow or brown in color?
 - (A) Carbon
 - (B) Lipofuscin
 - (C) Melanin
 - (D) Hemosiderin

48. Which of the following apoptic pathway is the main mechanism to eliminate self-recognizing lymphocyte?
- (A) Intrinsic
 - (B) Extrinsic
 - (C) Pathway related to ER stress
 - (D) None of above
49. Which kind of adaption is the main mechanism of barrettes esophagus?
- (A) Hyperplasia
 - (B) Hypertrophy
 - (C) Metaplasia
 - (D) Atrophy
50. Which of following is the main source of cytosolic calcium during cell injury and necrosis?
- (A) Extracellular fluid
 - (B) ER
 - (C) Mitochondria
 - (D) Golgi apparatus
51. Which of the following is not involved in extrinsic pathway of apoptosis?
- (A) FasL
 - (B) FADD
 - (C) Caspase 8
 - (D) Apoptosome

§ 謝明書老師—急性與慢性發炎（一）（二）

1. Infectious mononucleosis 在血液和淋巴腺裡出現的 atypical lymphocyte 是何種細胞？
 - (A) B cell
 - (B) $CD4^+$ T cell
 - (C) $CD8^+$ T cell
 - (D) Monocyte

2. Which of the following marks the right sequence of the extravasation events of leukocytes in acute inflammation?
 - (1) Transmigration
 - (2) Rolling
 - (3) Stasis and migration
 - (4) Chemotaxis
 - (5) Stable adhesion
 - (A) (3)(2)(5)(1)(4)
 - (B) (3)(1)(2)(4)(5)
 - (C) (2)(3)(5)(4)(1)
 - (D) (1)(2)(3)(4)(5)
 - (E) (3)(2)(5)(4)(1)

3. Which of the following statements regarding inflammation are correct?
 - (1) Endothelial contraction is the most common mechanism of increased vascular permeability.
 - (2) Granulomatous inflammation is a specific type of acute inflammation.
 - (3) Chronic inflammation is usually accompanied by tissue destruction and repair.
 - (4) Vasodilation is usually the earliest event of acute inflammation.
 - (5) Neutrophil is usually the first and predominant cell type to appear in acute inflammation.
 - (A) (1), (2), (3), (5)
 - (B) (1), (2), (4), (5)
 - (C) (2), (3), (4), (5)
 - (D) (1), (2), (3), (4)
 - (E) (1), (3), (4), (5)

4. Which of the following statements are correct?
- (1) C-C chemokine is the predominant chemotactic factor for neutrophil.
 - (2) TXA₂ is the major prostaglandin in endothelium.
 - (3) PGE₂ is related to fever in acute phase reaction.
 - (4) Substance P is related to transmission of pain signal.
 - (5) C5a is a strong chemotactic factor to neutrophil.
- (A) (2), (3)
(B) (3), (4), (5)
(C) (2), (3), (5)
(D) (1), (3), (5)
(E) (1), (2), (5)
5. (多選) Which of the following are true?
- (A) Gangrene refers to tissue necrosis with or without infection.
 - (B) Acute inflammation usually occurs around necrotic tissues.
 - (C) Apoptosis does not cause inflammation.
 - (D) Autolytic changes in the nucleus are pathognomonic of necrosis.
 - (E) The blood vessels surrounding an acutely inflamed site usually show progressive vasoconstriction from the time of injury.
6. (多選) 下列何者為組織間 transudate 增多的原因?
- (A) Liver cirrhosis
 - (B) Acute inflammation
 - (C) Nephrotic syndrome
 - (D) Right heart failure
 - (E) Hypothyroidism
7. Rolling adhesion 與下列何者無關?
- (A) P selectin
 - (B) Sialyl-Lewis X protein
 - (C) Integrin
 - (D) E selectin
8. Arachidonic acid metabolites (花生四烯酸衍生物) 對發炎反應非常重要，主要作用有 vessel dilation、vessel constriction、increased vascular permeability、chemotaxis 等。但有一種衍生物的功能卻是抑制發炎，請問是下列哪一種？
Which metabolite of arachidonic acid has anti-inflammatory effects?
- (A) Prostaglandins
 - (B) Leukotrienes
 - (C) Thromboxane A₂
 - (D) Lipoxins

9. 類固醇 (steroid) 能抑制發炎是因為它抑制了哪一個酵素，使 arachidonic acid metabolites 的合成降低？

Which enzyme playing a critical role in inflammation can be inhibited by steroids?

- (A) Cyclooxygenase
- (B) 5-lipoxygenase
- (C) Phospholipases
- (D) 12-lipoxygenase

10. 非類固醇消炎藥 (NSAIDs) 如阿斯匹靈 (aspirin) 能抑制發炎是因為它抑制了哪一種酵素，使 arachidonic acid metabolites 的合成降低？

Which enzyme playing a critical role in inflammation can be inhibited by NSAIDs?

- (A) Cyclooxygenase
- (B) 5-lipoxygenase
- (C) Phospholipases
- (D) 12-lipoxygenase

11. 下列何者可以活化 macrophages，對 TB 的免疫力極為重要？

- (A) IFN- γ
- (B) IL-1
- (C) TNF
- (D) NRAMPT 1

12. 發炎時，白血球要從血管進到組織，在 rolling 階段時相關的 adhesion molecules 哪一個是錯的？

Which adhesion molecule is not involved in rolling stage?

- (A) P selectin
- (B) Sialyl-Lewis X protein
- (C) Integrin
- (D) E selectin

13. 接續上題，白血球要從血管進到組織，在 adhesion 階段，不相關的 adhesion molecules 是哪一個？

- (A) Selectin
- (B) Integrin
- (C) ICAM-1
- (D) ICAM-2

14. 接續上題，白血球要從血管進到組織，在 transmigration 階段，最有相關的 adhesion molecules 是哪一個？
- (A) Selectin
 - (B) Integrin
 - (C) ICAM-1
 - (D) CD31
15. Cytokine 造成的 systemic effects 為何？
- (A) Fever
 - (B) Acute phase proteins
 - (C) Insulin resistance
 - (D) Leukocytosis
 - (E) 以上皆是 systemic effects
16. Chronic inflammation 的特色為何？
- (A) Mononuclear cells infiltration
 - (B) Repair and/or fibrosis/healing
 - (C) Tissue destruction
 - (D) 以上皆是
17. 下列何項不是 acute inflammation 病理學上的表現形態？
- (A) Suppurative inflammation
 - (B) Fibrinous inflammation
 - (C) Serous inflammation
 - (D) Granulomatous inflammation
18. 哪些敘述符合 neutrophil extracellular traps (NETs) ？
- (A) Neutrophil 的 antimicrobial mechanism 之一
 - (B) 成分為 chromatin 及 granule 中的 enzymes
 - (C) 與 autoimmune disease 和感染相關的 thrombotic disease (thrombotic microangiopathy) 有關聯性
 - (D) 細菌若帶有 DNase 則對 NETs 較有抗性
 - (E) 以上皆是
19. 哪一個 cytokine 跟組織修復與 connective tissue deposition 最相關？
- (A) IFN- γ
 - (B) TGF- β
 - (C) IL-1
 - (D) IL-17

20. Macrophage 如果接受到 IL-13 或 IL-4，較易形成哪種功能型態？
 - (A) M1 macrophage, associated with microbicidal actions and inflammation.
 - (B) M2 macrophage, associated with tissue repair, fibrosis and anti-inflammation.
21. Macrophage 如果接受到 IFN- γ ，較易形成哪種功能型態？
 - (A) M1 macrophage, associated with microbicidal actions and inflammation.
 - (B) M2 macrophage, associated with tissue repair, fibrosis and anti-inflammation.
22. 選項中哪一種微生物感染所引起的發炎反應與眾不同？
 - (A) Tuberculosis
 - (B) Leprosy
 - (C) Syphilis
 - (D) Streptococcus induced pneumonia
23. 哪一種 factor 在 anti-inflammation and repair 上有重要角色？
 - (A) IFN- γ
 - (B) TGF- α
 - (C) TGF- β
 - (D) TNF- α
24. 何者感染不以 granulomatous inflammation 表現？
 - (A) TB
 - (B) Leprosy
 - (C) Community acquired pneumonia
 - (D) Cat scratch disease
25. Vasodilation、permeability 上升、edema、fibrin 累積是下列何者特徵？
 - (A) 急性發炎
 - (B) 慢性發炎
 - (C) 乾性壞疽
 - (D) 肉芽腫性發炎
26. 下列哪一個黏著分子（adhesion molecules）能存在於白血球上，使得白血球可以強力黏附於血管內皮細胞（blood vessel endothelial）上？
 - (A) E selectin
 - (B) ICAM 1
 - (C) Integrins
 - (D) Fibronectin

27. 巨噬細胞和嗜中性球是體內主要吞噬細胞，下列何者不是吞噬細胞用來殺死病原的相關分子？
- (A) O_2^-
 - (B) NO
 - (C) OCI^-
 - (D) HCO_3^-
28. 有一個人的白血球游動能力正常，吞噬功能也正常，但卻無法殺掉吞噬的細菌，請問他可能有哪種疾病？
- (A) Leukocyte adhesion deficiency
 - (B) Chronic granulomatous
 - (C) Wiskott Aldrich syndrome
 - (D) Di George syndrome
29. 以下發炎反應與相關 factors 的配對何者錯誤？
- (A) Vasodilation: TNF, IL-1
 - (B) Chemotaxis: c5a
 - (C) Pain: Prostaglandins
 - (D) Fever: Bradykinin
30. 下列有關急性炎症病理的敘述，何者錯誤？
- (A) 小血管收縮
 - (B) 血流速度變慢
 - (C) 白血球聚集
 - (D) 血管外液體增加
31. 下列何種炎症的滲出物僅含有少量細胞（poor-cell fluid）？
- (A) 肉芽性
 - (B) 出血性
 - (C) 漿液性
 - (D) 化膿性
32. 化膿性滲出物最可能出現何物質？
- (A) 嗜中性球和壞死組織
 - (B) 嗜中性球和黏液
 - (C) 巨噬細胞和黏液
 - (D) 壞死組織和結締組織

§ 謝明書老師—組織修復（一）（二）

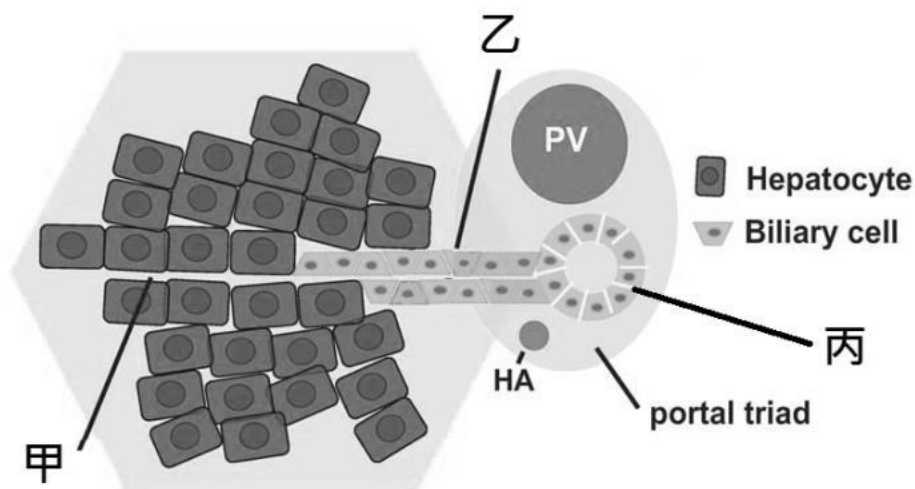
1. Which of the following statements regarding inflammation are correct?

- (1) Endothelial contraction is the most common mechanism of increased vascular permeability.
- (2) Granulomatous inflammation is a specific type of acute inflammation.
- (3) Chronic inflammation is usually accompanied by tissue destruction and repair.
- (4) Vasodilation is usually the earliest event of acute inflammation.
- (5) Neutrophil is usually the first and predominant cell type to appear in acute inflammation.

- (A) (1), (2), (3), (5)
- (B) (1), (2), (4), (5)
- (C) (2), (3), (4), (5)
- (D) (1), (2), (3), (4)
- (E) (1), (3), (4), (5)

2. （多選）下列描述，何者正確？

- (A) M2 macrophage 分泌 transforming growth factor- β 促進 fibroblast 聚集增生
- (B) 如下圖，肝臟的幹細胞通常位於箭頭甲所指的地方



- (C) Healing of skin by first retention 較 secondary retention 不需要 myofibroblast，且可以修復汗腺、毛囊。由此可知，淺二度燒傷的皮膚可以保留完整排汗功能
- (D) 分子 CXCR 3、IL-6 可促進皮膚的 epithelial cells 新生
- (E) 當我們取 keloid 做成組織玻片，並以 Masson's trichrome stain 染色，可以發現 epidermis 有不正常增厚的角質堆積

3. 關於 stem cell 的敘述下列何者錯誤？
- (A) Totipotency is the ability of a single cell to divide and produce all the differentiated cells in an organism, including extra embryonic tissues.
 - (B) Pluripotency refers to a stem cell that has the potential to differentiate into any of the three germ layers, like a zygote.
 - (C) Multipotency stem (progenitors) cells have the potential to give rise to cells from multiple, but a limited number of lineages.
 - (D) Pluripotent stem cells can give rise to any fetal or adult cell type.
4. Which of the following statements about description related to growth factor and their receptors is incorrect?
- (A) EGFR1 mutations and amplification have been detected in cancers of the lung, head and neck, and breast, glioblastomas, and other cancers.
 - (B) TGF- α has homology with EGF, binds to EGFR, and shares most of the biologic activities of EGF.
 - (C) VEGFR2, located in endothelial cells and many other cell types, is the main receptor for the vasculogenic and angiogenesis effects of VEGF.
 - (D) VEGF-C and VEGF-D bind to VEGFR-1 and act on lymphatic endothelial cells to induce the production of lymphatic vessels (lymphangiogenesis).
5. Which of following statements about angiogenesis is incorrect?
- (A) The mechanism of angiogenesis includes angiogenesis from preexisting vessels and angiogenesis from epithelial precursors cells.
 - (B) VEGF include delta-like ligand 4 in stalk cells, while Notch 1 and Notch 4 are expressed in tip cells.
 - (C) VEGF stimulates delta-like ligand 4 (Dll4) / Notch, which inhibits VEGFR signaling.
 - (D) Dll4 blockade causes an increase in capillary sprouting and endothelial cell (EC) proliferation, creating vessels that are disorganized and have a small lumen size.
 - (E) VEGF blockade decreases capillary sprouting, and the proliferation and survival of ECs.
6. 下列何者可以活化 macrophages，對 TB 的免疫力極為重要？
- (A) IFN- γ
 - (B) IL-1
 - (C) TNF
 - (D) NRAMP1

7. 人體組織受損後，可經由再生（**regeneration**）或修補（**repair**）兩種方式來復原。下列關於再生與修補的敘述中，何者有誤？
- (A) 完全以再生方式復原的器官，功能可恢復至接近受損前的狀態
 - (B) 即使是具有再生能力的器官（如肝臟），如果細胞外間質（**extracellular matrix**，**ECM**）嚴重受損，也無法完全以再生方式復原
 - (C) 不論是再生或修補，過程中都會牽涉細胞的增生
 - (D) 成人肝臟的再生，主要是靠原生細胞（**blastemal cells**）重新分裂、分化來補充缺少的成熟肝細胞
8. 組織再生能力常與幹細胞的功能有關，因此對幹細胞的了解是再生醫學的基礎。關於幹細胞的敘述，何者有誤？
- (A) 胚胎幹細胞（**embryonal stem cells**，**ES cells**）來自囊胚（**blastocyst**）的內細胞團塊（**inner cell mass**），具有分化為三胚層細胞的能力
 - (B) 相對地，成人幹細胞（**adult stem cells**）的分化能力較為有限，通常只能分化成少數種類的細胞
 - (C) 在成人體內，大多數器官（如心臟與腦部）並不存在幹細胞，因此再生能力相當有限
 - (D) 將特殊基因（如 **Oct3/4**、**Sox2**、**c-myc** 與 **Klf4** 四基因的組合）置入已分化之體細胞核內，使其表現，可誘導其轉變為幹細胞，稱為 **iPS 細胞**（**induced pluripotent stem cells**）
9. 關於細胞週期（**cell cycle**）與細胞增生的敘述，何者有誤？
- (A) 成人體內部份細胞（如肝細胞）平時處在不分裂的 **G0 phase**，但有需要時可以進入 **G1 phase** 而進行有絲分裂
 - (B) 上述細胞由 **G0 phase** 進入 **G1 phase**，主要是靠某些訊息（如生長因子、荷爾蒙等）的刺激，啟動增生相關的基因
 - (C) 在這些訊息中，人類表皮生長因子（**human epidermal growth factor**）主要是跟細胞核內的受體（**receptor**）結合，啟動細胞內的後續反應
 - (D) **Integrin** 等結合性醣蛋白（**adhesive glycoprotein**）也能啟動細胞內的訊息傳遞機制，促進細胞增生
10. 膠原蛋白質（**collagen**）是構成細胞外間質（**extracellular matrix**，**ECM**）的主要成分之一。關於膠原蛋白質的敘述，何者有誤？（範圍未知）
- (A) 基本結構為雙股螺旋形之蛋白質
 - (B) 膠原蛋白質的合成需要維生素 C（**vitamin C**）
 - (C) 第一型膠原蛋白質會平行排列、聚合成繩索狀，在真皮等結締組織提供抗張能力
 - (D) 第四型膠原蛋白質為基底膜（**basement membrane**）主成分之一

11. 關於急性傷害後的組織修補，何者有誤？

- (A) 以皮膚切割傷的癒合 (healing) 為例，大致上可分為發炎、增生、成熟三個階段
- (B) 在發炎時期，巨噬細胞 (macrophage) 扮演承先啟後的腳色；他一方面可清除傷口的組織碎屑與細菌，另一方面也能放出訊息分子，促進血管新生與其他細胞增生
- (C) 增生的肉芽組織 (granulation tissue) 中有大量纖維母細胞 (fibroblasts)，因此在這個時期，細胞外間質已堆積緻密的膠原纖維
- (D) 除了纖維母細胞增加以外，血管的增生也是肉芽組織的特徵之一

12. 關於 angiogenesis 的敘述，何者有誤？

- (A) Angiogenesis 指「由既有血管長出新的血管」，與 vasculogenesis 是同義詞
- (B) 在 angiogenesis 的過程中，血管內皮細胞不僅要增生，也需要移動到適當位置以形成新血管
- (C) 顧名思義，血管內皮生長因子 (vascular endothelial growth factor, VEGF) 可促進血管內皮細胞增生及移動，在 angiogenesis 扮演重要的角色
- (D) 除了物理性傷害，組織缺氧也會引發 angiogenesis

13. 皮膚傷口的癒合在臨床上相當常見。關於其相關敘述，何者有誤？

- (A) 傷口範圍過大，或是傷口髒污未清除時，會延緩癒合的進行
- (B) 類固醇藥物會抑制發炎，使傷口提前進入成熟期，加速癒合的進行
- (C) 外科手術切割傷的傷口若經適當縫合，未受感染，經過十天左右表皮即可大致復原，但此時真皮的疤痕 (scar) 組織尚未完全成熟，若用力拉扯或壓力劇烈改變，傷口就有裂開的危險
- (D) 承上，一般而言，術後三個月時疤痕已完全成熟，此時組織中的膠原纖維已緻密排列，相對地纖維母細胞不多

14. 關於傷口癒合與組織修補相關病變的敘述，何者有誤？

- (A) 在皮膚切割傷，肉芽組織形成過量時會妨礙表皮復原，延緩癒合的進行
- (B) 蟹足腫 (keloid) 是癒合過程中，纖維母細胞過度增生轉變而成的良性腫瘤。顯微鏡檢查可發現大量良性纖維母細胞，以及少量膠原纖維
- (C) 大面積燒燙傷的患者，可能會因疤痕收縮過度而使肢端變形
- (D) 若發炎持續進行，組織內膠原纖維不斷累積，就會造成纖維化 (fibrosis)，嚴重到一定程度時會妨礙器官功能。國人常見的肝硬化 (liver cirrhosis) 即為一例

15. Cytokine 造成的 systemic effects 為何？

- (A) Fever
- (B) Acute phase proteins
- (C) Insulin resistance
- (D) Leukocytosis
- (E) 以上皆是 systemic effects

16. Chronic inflammation 的特色為何？
 - (A) Mononuclear cells infiltration
 - (B) Repair and/or fibrosis/healing
 - (C) Tissue destruction
 - (D) 以上皆是
17. Based on proliferative activity, the tissues of the body are divided into labile tissues, stable tissues, and permanent tissues. Which of following statements about description related to this classification is incorrect?
 - (A) Hematopoietic cells in the bone marrow and the majority of surface epithelia are labile tissues.
 - (B) All the stable tissues have a limited capacity to regenerate after injury.
 - (C) The majority of neurons and cardiac muscle cells are permanent tissues.
 - (D) In permanent tissues, repair is typically dominated by scar formation. repair and/or fibrosis/healing tissue destruction
18. Which of following statements about healing of skin wounds and fibrosis in parenchymal organs is incorrect?
 - (A) The basic mechanisms of fibrosis are the same as those of scar formation in the skin during tissue repair.
 - (B) TGF- β is the most important involving cytokine for the synthesis and deposition of connective tissue proteins.
 - (C) In all internal organs, fibroblasts are the main source of collagen.
 - (D) In healing of skin wounds, wound contraction is performed by the action of myofibroblasts.
 - (E) In healing of skin wounds, the dermal appendages destroyed in the line of the incision are permanently lost.
19. 哪一個 cytokine 跟組織修復與 connective tissue deposition 最相關？
 - (A) IFN- γ
 - (B) TGF- β
 - (C) IL-1
 - (D) IL-17
20. Macrophage 如果接受到 IL-13 或 IL-4，較易形成哪種功能型態？
 - (A) M1 macrophage, associated with microbicidal actions and inflammation.
 - (B) M2 macrophage, associated with tissue repair, fibrosis and anti-inflammation.

21. Macrophage 如果接受到 IFN- γ ，較易形成哪種功能型態？
- (A) M1 macrophage, associated with microbicidal actions and inflammation.
 - (B) M2 macrophage, associated with tissue repair, fibrosis and anti-inflammation.
22. 胚胎中的哪一群細胞是 pluripotent cells，又稱為 embryonic stem cells？
- (A) Zygote
 - (B) Inner cell mass of blastocyst
 - (C) Placenta
 - (D) Yolk sac
23. Angiogenesis 中，tip cells 與 stalk cells 的分化是由哪一個 signal pathway 所誘發？
- (A) EGFR
 - (B) Notch signaling
 - (C) Wnt/ β -catenin
 - (D) MAPK/ERK pathway
24. Repair 中過度的 collagen 及 connective tissue deposition，會造成下列哪些情況？
- (A) Hypertrophic scar
 - (B) Keloid
 - (C) Contraction
 - (D) 以上皆是
25. Which function is not related to lncRNA? (範圍未知)
- (A) Gene editing
 - (B) Gene suppression
 - (C) Promote chromatin modification
 - (D) Assembly of protein complexes
26. Which of the followings is not intermediate filament? (範圍未知)
- (A) Desmin
 - (B) Glial fibrillary acidic protein (GFAP)
 - (C) Cytokeratin
 - (D) Smooth muscle actin
27. What is Warburg effect? (範圍未知)
- (A) Energy transfer between cells
 - (B) Programmed cell death after nutritional deprivation
 - (C) Metabolic switch in cancers
 - (D) Hormone changes after glucose intake

28. 選項中哪一種微生物感染引起的發炎反應與眾不同？（範圍未知）

- (A) Tuberculosis
- (B) Leprosy
- (C) Syphilis
- (D) Streptococcus induced pneumonia

29. 哪一種 factor 在 anti-inflammation and repair 上有重要角色？

- (A) IFN- γ
- (B) TGF- α
- (C) TGF- β
- (D) TNF- α

30. 何者感染不以 granulomatous inflammation 表現？

- (A) TB
- (B) Leprosy
- (C) Community acquired pneumonia
- (D) Cat scratch disease

§ 連晃駿老師—血行動力學病理（一）（二）（三）

1. （多選）Which of the following is not risk factor for embolism?
 - (A) Disseminated cancer
 - (B) Prothrombin mutation
 - (C) Bernard-Soulier syndrome
 - (D) Antiphospholipid antibody syndrome
 - (E) Long-bone fracture
 - (F) Trousseau syndrome
 - (G) Leiden mutation
 - (H) von-Willebrand disease
 - (I) Decompression sickness
2. （多選）The following conditions may cause central retinal vein occlusion in a
3. young patient?
 - (A) Protein C deficiency
 - (B) Excess protein S
 - (C) Antithrombin III deficiency
 - (D) Atrial fibrillation
 - (E) Factor V Leiden mutation
4. （多選）Which of the following are false about cerebral infarction?
 - (A) The area of infarct tends to be wedge-shaped.
 - (B) It can result from thrombosis of the external carotid artery disease.
 - (C) Liquefactive necrosis occurs in the brain tissue.
 - (D) Cortical blindness can result from infarction of the anterior cerebral artery.
 - (E) The necrotic tissue is eventually replaced by Schwann cells.
5. （多選）下列何者為組織間 transudate 增多的原因？
 - (A) Liver cirrhosis
 - (B) Acute inflammation
 - (C) Nephrotic syndrome
 - (D) Right heart failure
 - (E) Hypothyroidism
6. Hypoxia, strictly defined, means:
 - (A) Failure of intracellular oxidative phosphorylation for any reason
 - (B) Too little perfusion of the tissues
 - (C) Too little oxygen reaching the mitochondria
 - (D) Too little oxygen in the lungs
 - (E) Too little oxygen obtained by an organism

7. Which of the following factor have anti-thrombotic activity?
 - (1) ADPase
 - (2) Thrombomodulin
 - (3) Tissue factor pathway inhibitor
 - (4) α 2-antiplasmin
 - (5) Plasminogen activator (PA)
 - (A) (1), (2), (3), (5)
 - (B) (2), (3), (4), (5)
 - (C) (1), (2), (3), (4)
 - (D) (1), (3), (4), (5)
 - (E) (1), (2), (4), (5)
8. Which of the following factor have anti-thrombotic activity?
 - (1) Protein C
 - (2) Thrombomodulin
 - (3) Tissue factor pathway inhibitor
 - (4) Heparin-like molecules
 - (5) Plasminogen activator inhibitor (PAI)
 - (A) (1), (2), (3), (4)
 - (B) (2), (3), (4), (5)
 - (C) (1), (2), (3), (5)
 - (D) (1), (3), (4), (5)
 - (E) (1), (2), (3), (4), (5)
9. Which of the following is not usually found in patient with shock?
 - (A) Ischemic encephalopathy
 - (B) Diffuse alveolar damage
 - (C) Acute tubular necrosis
 - (D) Metabolic alkalosis
 - (E) All of the above

10. Which of the followings is commonly associated with shock?

- (1) Red neuron
- (2) Diffuse alveolar damage
- (3) Acute tubular necrosis
- (4) Metabolic alkalosis
- (5) Ischemic necrosis of intestine
- (6) Centrizonal necrosis of liver
- (7) Ischemic encephalopathy
- (A) (1), (2), (3), (4)
- (B) (1), (3), (4), (5)
- (C) (2), (3), (4), (5), (6)
- (D) (1), (2), (3), (5), (6), (7)
- (E) (1), (2), (4), (5), (6), (7)

11. Which of the followings are related to pathogenesis of septic shock?

- (1) Decreased protein C
- (2) DIC
- (3) Vasodilation
- (4) Myocardial depression
- (5) Hypoglycemia
- (A) (1), (2), (3), (4)
- (B) (2), (3), (4), (5)
- (C) (1), (2), (3), (5)
- (D) (1), (3), (4), (5)
- (E) (1), (2), (4), (5)

12. Which of the followings are related to pathogenesis of septic shock?
- (1) Increased anti-coagulant
 - (2) Decreased plasminogen activator inhibitor
 - (3) Adrenal insufficiency
 - (4) Hypoglycemia
 - (5) Insulin resistance
 - (6) DIC
- (A) (1), (2), (3), (4)
 - (B) (2), (3), (4), (5)
 - (C) (1), (2), (3), (5)
 - (D) (3), (5), (6)
 - (E) (1), (2), (4)
13. Which of the following is/are prone to result in red infarct?
- (1) Pulmonary infarction
 - (2) Renal infarction
 - (3) Splenic infarction
 - (4) Reperfusion of myocardial infarction
 - (5) Torsion of testis
 - (6) Ovarian torsion
- (A) (1), (2), (3)
 - (B) (2), (3), (4)
 - (C) (1), (2), (5)
 - (D) (3), (4), (5)
 - (E) (1), (4), (5), (6)

14. Which of the following are common symptoms and signs seen in a patient with left heart failure due to left-sided myocardial infarction?

- (1) Vegetations of mitral valve
- (2) Pulmonary edema
- (3) Nutmeg liver
- (4) Dyspnea
- (5) Lower leg edema
- (6) Brown induration of lung
- (7) Air hunger
- (8) Increased central venous pressure
- (A) (1), (2)
- (B) (3), (4)
- (C) (2), (3)
- (D) (1), (3)
- (E) (2), (4), (6), (7)

15. Which of the following finding(s) is found in patients with right heart failure?

- (1) Centrilobular congestion/necrosis of liver
- (2) Pulmonary edema
- (3) Lymphedema
- (4) Dyspnea
- (5) Increased central venous pressure
- (6) Brown induration of lung
- (7) Nutmeg liver
- (8) Lower leg edema
- (A) (1), (5), (7), (8)
- (B) (3), (4)
- (C) (2), (3)
- (D) (1), (3)
- (E) (2), (4)

16. Which of the following is/are associated with hypercoagulability?

- (1) Migratory thrombophlebitis
- (2) Long-term bed ridden
- (3) Leiden mutation
- (4) Bernard Soulier syndrome
- (5) Anti-phospholipid syndrome
- (A) (1), (2), (3), (4)
- (B) (2), (3), (4), (5)
- (C) (1), (2), (3), (5)
- (D) (1), (3), (4), (5)
- (E) (1), (2), (3), (4), (5)

17. Which of the following is/are associated with hypercoagulability?

- (1) von Willebrand disease
- (2) Trousseau syndrome
- (3) Prothrombin mutation (3'-UTR transition in prothrombin gene)
- (4) Heparin-induced thrombocytopenia
- (5) Anti-phospholipid syndrome
- (6) Congestive heart failure
- (7) Prosthetic cardiac valve
- (8) Atherosclerosis
- (9) Disseminated cancer
- (A) (1), (2), (3), (4)
- (B) (1), (3), (4), (5), (6)
- (C) (1), (2), (3), (5), (7), (8)
- (D) (2), (3), (4), (5), (6), (7), (8), (9)
- (E) (1), (2), (3), (4), (5), (7), (8), (9)

18. Which of the following factor have anti-thrombotic activity?

- (1) Protein C
- (2) Thrombomodulin
- (3) Tissue factor pathway inhibitor
- (4) Heparin-like molecules
- (5) Plasminogen activator inhibitor (PAI)
- (6) Tissue factor
- (A) (1), (2), (3), (4)
- (B) (2), (3), (4), (5)
- (C) (1), (2), (3), (5)
- (D) (1), (3), (4), (5)
- (E) (1), (2), (3), (4), (5)

19. Which of the following are potential outcomes of thrombosis?

- (1) White infarct
- (2) Liquefactive necrosis
- (3) Myocardial infraction
- (4) Pulmonary embolism
- (5) Red infarct
- (6) Caseous necrosis
- (7) Stroke
- (8) Acute cor pulmonale
- (A) (1), (2)
- (B) (3), (4)
- (C) (1), (3), (5)
- (D) (1), (2), (3), (5), (7)
- (E) (1), (2), (3), (4), (5), (7), (8)

20. Which of the following are potential outcomes of embolism?

- (1) White infarct
- (2) Caisson disease
- (3) Coagulative necrosis
- (4) Caseous necrosis
- (5) Acute cor pulmonale
- (A) (2), (3), (4), (5)
- (B) (1), (3), (4), (5)
- (C) (1), (2), (3), (5)
- (D) (1), (2), (3), (4)
- (E) (1), (2), (3), (4), (5)

21. Which of the following statements are right?

- (1) Septic shock can be caused by Gram-positive bacterial infection.
- (2) Cardiogenic shock with large myocardial infarction usually carries poor prognosis.
- (3) DIC may result in consumption coagulopathy (and bleeding tendency).
- (4) Venous thrombi are usually caused by endothelial injury.
- (5) Cool and pale skin is an early sign of a septic shock.
- (6) Cardiac thrombi are usually related to endothelial/endocardial injury.
- (A) (1), (2), (4)
- (B) (1), (4), (5), (6)
- (C) (1), (2), (3), (6)
- (D) (2), (3), (5)
- (E) (2), (3), (4)

22. Which of the following statements are right?

- (1) Hypoglycemia is a common systemic effect of shock associated with systemic infection.
- (2) Cardiogenic shock with large myocardial infarction usually carries poor prognosis.
- (3) DIC may result in bleeding tendency.
- (4) Endothelial injury is a common cause of thrombosis in arterial and cardiac system.
- (5) Cool and pale skin is an early sign of severe hypovolemic shock.
- (A) (1), (2), (3), (4)
- (B) (1), (3), (4), (5)
- (C) (1), (2), (4), (5)
- (D) (2), (3), (4), (5)
- (E) (1), (2), (3), (4), (5)

23. Which of the following is not related to edema?
- (A) Filariasis-associated elephantiasis
 - (B) Severe malnutrition
 - (C) Increased osmotic pressure
 - (D) Nephritic syndrome
 - (E) Right heart failure
24. Which of the following is not related to edema?
- (A) Elephantiasis
 - (B) Severe malnutrition
 - (C) Increased osmotic pressure
 - (D) Protein-losing gastroenteropathy
 - (E) Acute inflammation
25. Which of the following is usually not present in a patient with advanced liver cirrhosis?
- (A) Hyperalbuminemia
 - (B) Edema
 - (C) Congestive splenomegaly
 - (D) Decreased osmotic pressure
 - (E) Ascites

B05 第 1 次區段考

1. Which of the following is not part of practice anatomical pathology?
 - (A) Cytopathology
 - (B) Medical genetics
 - (C) Forensic pathology
 - (D) Surgical pathology
2. The final pathological of full body postmortem examination is:
 - (A) DNA genomic testing
 - (B) Autopsy
 - (C) 3D reconstruction of gross surgical specimen
 - (D) Digital pathological examination
3. A board-certified physician can perform autopsy at his clinical bath tub?
 - (A) Yes, he can and is well-done.
 - (B) Yes, he can and needs a pathologist's assistance.
 - (C) he cannot and is illegal.
 - (D) No, he cannot except to report to the in-charge district prosecutor.
4. “See for oneself” is a described method of?
 - (A) Autopsy
 - (B) Medical ethic
 - (C) Medical esthetics
 - (D) Medical esthesiography
5. What is the major cause of the decline of the autopsy rate?
 - (A) Religionism
 - (B) Poor autopsy technique
 - (C) Few research value based on autopsy
 - (D) No interest in physicians and pathologists
6. Which kind of adaption is the main mechanism of barrettes esophagus?
 - (A) Hyperplasia
 - (B) Hypertrophy
 - (C) Metaplasia
 - (D) Atrophy

7. Which of following is the main source of cytosolic calcium during cell injury and necrosis?
 - (A) Extracellular fluid
 - (B) ER
 - (C) Mitochondria
 - (D) Golgi apparatus
8. Which of the following is not involved in extrinsic pathway of apoptosis?
 - (A) FasL
 - (B) FADD
 - (C) Caspase 8
 - (D) Apoptosome
9. Which of the following molecules is most important for macroautophagy?
 - (A) Necrosome
 - (B) TNFR1
 - (C) LC3
 - (D) Chaperone
10. Vasodilation、permeability 上升、edema、fibrin 累積是下列何者特徵？
 - (A) 急性發炎
 - (B) 慢性發炎
 - (C) 乾性壞疽
 - (D) 肉芽腫性發炎
11. 下列哪一個黏著分子（adhesion molecules）能存在於白血球上，使得白血球可以強力黏附於血管內皮細胞（blood vessel endothelial）上？
 - (A) E selectin
 - (B) ICAM 1
 - (C) Integrins
 - (D) Fibronectin
12. 巨噬細胞和嗜中性球是體內主要吞噬細胞，下列何者不是吞噬細胞用來殺死病原的相關分子？
 - (A) O_2^-
 - (B) NO
 - (C) OCI^-
 - (D) HCO_3^-

13. 有一個人的白血球游動能力正常，吞噬功能也正常，但卻無法殺掉吞噬的細菌，請問他可能有哪種疾病？
 - (A) Leukocyte adhesion deficiency
 - (B) Chronic granulomatous
 - (C) Wiskott Aldrich syndrome
 - (D) Di George syndrome
14. 以下發炎反應與相關 factors 的配對何者錯誤？
 - (A) Vasodilation: TNF, IL-1
 - (B) Chemotaxis: c5a
 - (C) Pain: Prostaglandins(pg)
 - (D) Fever: Bradykinin
15. 下列有關急性炎症病理的敘述，何者錯誤？
 - (A) 小血管收縮
 - (B) 血流速度變慢
 - (C) 白血球聚集
 - (D) 血管外液體增加
16. 下列何種炎症的滲出物僅含有少量細胞（poor-cell fluid）？
 - (A) 肉芽性
 - (B) 出血性
 - (C) 漿液性
 - (D) 化膿性
17. Which of the following are associated with hypercoagulability?
 - (1) Atherosclerosis
 - (2) Disseminated cancer
 - (3) 3-UTR transition in Prothombin gene
 - (4) Anti-phospholipid antibody syndrome
 - (5) von-Willbrand disease
 - (A) (2), (3), (4), (5)
 - (B) (1), (3), (4), (5)
 - (C) (1), (2), (3), (5)
 - (D) (1), (2), (3), (4)
 - (E) (1), (2), (3), (4), (5)

18. Which of the followings are common symptoms and signs in a patient with right heart failure?
- (1) Increased central venous pressure
 - (2) Brown in duration of lung
 - (3) Nutmeg liver
 - (4) Dyspnea
 - (5) Lower leg edema
- (A) (1), (2), (3)
(B) (3), (5)
(C) (1), (2), (5)
(D) (1), (3), (5)
(E) (2), (3), (4)
19. Which of the followings are findings associated with thromboembolism?
- (1) White infarct
 - (2) Caseous necrosis
 - (3) Coagulative necrosis
 - (4) Caisson disease
 - (5) Liquefactive necrosis
- (A) (1), (2), (3), (5)
(B) (3), (4), (5)
(C) (1), (3), (4), (5)
(D) (1), (3), (5)
(E) (1), (2), (3), (4), (5)
20. Which of the following factors are associated with anti-thrombotic activity?
- (1) Protein C
 - (2) Thrombomodulin
 - (3) Tissue factor
 - (4) t-PA
 - (5) Heparin-like molecule
- (A) (1), (2), (4)
(B) (1), (2), (3), (4)
(C) (1), (2), (3), (4), (5)
(D) (1), (2), (4), (5)

21. Which of the following pathological findings can be common findings associated with shock?

- (1) Ischemic encephalopathy
- (2) Diffuse alveolar damage
- (3) Acute tubular necrosis
- (4) Metabolic alkalosis
- (5) Centrizonal necrosis liver

(A) (1), (2), (3), (5)

(B) (1), (3), (4), (5)

(C) (2), (3), (4), (5)

(D) (1), (2), (4), (5)

(E) (1), (2), (3), (4), (5)

22. Which of the following statements are right?

- (1) Hyperglycemia is a common systemic effect of shock associated with systemic inflammation.
- (2) Cardiogenic shock with large myocardial infarction usually carries poor prognosis.
- (3) DIC may result in bleeding tendency.
- (4) Endothelial injury is a common cause of thrombosis in arterial and cardiac system.
- (5) Cool and pale skin is an early sign of severe hypovolemic shock.

(A) (1), (2), (3), (4)

(B) (1), (3), (4), (5)

(C) (1), (2), (3), (5)

(D) (2), (3), (4), (5)

(E) (1), (2), (3), (4), (5)

23. Which of the followings can be risk factor for embolism?

- (1) Bernard-Soulier syndrome
- (2) Leiden mutation
- (3) Anti-phospholipid syndrome
- (4) Decompression disease
- (5) Long bone fracture

(A) (1), (2), (3), (4)

(B) (1), (3), (4), (5)

(C) (1), (2), (3), (5)

(D) (2), (3), (4), (5)

(E) (1), (2), (3), (4), (5)

24. Which of the following is prone to result in red infarct?

- (1) Pulmonary embolism
 - (2) Renal infarction
 - (3) Ovary torsion
 - (4) Reperfusion of myocardial infarction
 - (5) Intestinal infarction
- (A) (1), (2), (3), (4)
 - (B) (1), (2), (4), (5)
 - (C) (2), (3), (4), (5)
 - (D) (1), (2), (3), (4), (5)
 - (E) (1), (3), (4), (5)

B06 第 1 次區段考

1. What is the method most commonly used to detect gene amplification in cancer?
 - (A) Immunochemistry
 - (B) Florescent in situ hybridization (FISH)
 - (C) Chromogen in situ hybridization
 - (D) Sanger
2. Which is most likely etiology of gastric lymphoma?
 - (A) Epstein-Barr virus
 - (B) Helicobacter pylori
 - (C) Nitrosamine
 - (D) Ractopamin
3. Which one needs respiratory protection during autopsy?
 - (A) 缺
 - (B) AIDS patient
 - (C) Covid 19 patient
 - (D) Patient with bacterial pneumonia
4. The purpose of hospital autopsy is based on:
 - (A) Doctor's academic interest
 - (B) Understanding of diseases
 - (C) Laboratory drug test
 - (D) Clinical experiment
5. What does autopsy mean for?
 - (A) Necropsy
 - (B) Postmortem dissection
 - (C) Dissection of organs
 - (D) See for oneself
6. The hospital autopsy needs:
 - (A) Consent by oneself or family member
 - (B) Physician's consent
 - (C) Treatment drugs
 - (D) Professor at large

7. Which of the following is not a feature of ferroptosis?
- (A) Iron
 - (B) Increased ROS
 - (C) Non-lytic cell death
 - (D) Inflammation
8. Which of the following descriptions about autophagy is wrong?
- (A) Cancer cells may increase autophagy.
 - (B) Some neurodegenerative diseases are caused by insufficient autophagy.
 - (C) Autophagy is the main mechanism for recycling misfolded proteins.
 - (D) Autophagosome is double membrane structure.
9. Which one can be recognized by Annexin V of Annexin apoptotic kit?
- (A) Propidium iodide
 - (B) Phosphatidylserine
 - (C) DAMP
 - (D) DNA
10. 下列何者和 tuberculosis 有關？
- (A) Caseous necrosis
11. Which of the following can be considered as irreversible cell injury rather than reversible cell injury?
- (A) Plasma membrane rupture
 - (B) Cytoplasmic eosinophilia
 - (C) Myelin figure
 - (D) Cellular swelling
12. Steatosis 是什麼的累積？
- (A) 缺
 - (B) Lipofuscin
 - (C) Triglyceride
 - (D) Cholesterol
13. 白血球從血管進到組織所需之各項分子配對，何者正確？
- (A) Rolling：selectin 和 sialyl-Lewis x glycoprotein
 - (B) Adhesion：I-CAM 和 selectin
 - (C) Transmigration：CD31 和 CD31
 - (D) Chemotaxis：LPS 和 G protein-coupled receptor
 - (E) 以上皆是

14. NSAID 會抑制下列哪個合成 arachidonic acid metabolite 所需的酵素？
 - (A) 5-lipoxygenase
 - (B) Phospholipase
 - (C) Cyclooxygenase
 - (D) Lipoxins
15. 缺
 - (A) Cyclooxygenase
16. 哪些敘述符合 neutrophil extracellular traps (NETs) ？
 - (A) Neutrophil 的 antimicrobial mechanism 之一
 - (B) 成分為 chromatin 及 granule 中的 enzymes
 - (C) 與 autoimmune disease 和感染相關的 thrombotic disease (thrombotic microangiopathy) 有關聯性
 - (D) 細菌若帶有 DNase 則對 NETs 較有抗性
 - (E) 以上皆是
17. 缺
 - (A) 缺
 - (B) 缺
 - (C) 缺
 - (D) 缺
 - (E) 以上皆是 systemic effects
18. 下列何項不是 acute inflammation 病理學上的表現形態？
 - (A) Suppurative inflammation
 - (B) Fibrinous inflammation
 - (C) Serous inflammation
 - (D) Granulomatous inflammation
19. Chronic inflammation 的特色為何？
 - (A) Mononuclear cells infiltration
 - (B) Repair and fibrosis
 - (C) Tissue destruction
 - (D) 以上皆是
20. 下列有關 macrophage 的敘述何者正確？
 - (A) Macrophage 接收到 IL-13 或 IL-4 會形成 M1 macrophage
 - (B) Macrophage 接收到 IFN- γ 會形成 M2 macrophage
 - (C) M1macrophage is associated with microbicidal actions and inflammation.
 - (D) M1macrophage 活化其他細胞主要透過 TGF- β

21. 下列病原體何者引起的發炎反應不同？

- (A) 缺
- (B) *Streptococcus pneumonia*

22. 缺

- (A) 缺
- (B) Dll-4 and Notch signaling

23. Which of the following finding(s) is found in patients with right heart failure?

- (1) Dyspnea
 - (2) Brain edema
 - (3) Centrilobular congestion
 - (4) Increased central venous pressure
 - (5) Pitting edema
- (A) (3), (4), (5)
 - (B) (3), (4)
 - (C) (1), (4), (5)
 - (D) (1), (3), (5)
 - (E) (2), (3), (4)

24. 下列何者和 hypercoagulability 有關？

- (1) Prothrombin G20210A mutation
 - (2) Trousseau syndrome
 - (3) Congestive heart failure
 - (4) Bernard-Soulier syndrome
 - (5) von Willebrand syndrome
- (A) (1), (2), (3)
 - (B) (2), (3), (4), (5)
 - (C) (1), (2), (3), (5)
 - (D) (1), (2), (3), (4)
 - (E) (1), (2), (3), (4), (5)

25. Which of the following factors are associated with anti-thrombotic activity?
- (1) Protien C
 - (2) Thrombomodulin
 - (3) Tissue factor
 - (4) t-PA
 - (5) Herparin-like molecule
 - (A) (1), (2), (4)
 - (B) (1), (2), (3), (4)
 - (C) (1), (2), (3), (4), (5)
 - (D) (1), (2), (4), (5)
26. 缺，推測為(Which of the following is related to edema?)
- (A) Elephantiasis
 - (B) Excessive sodium intake with renal insufficiency
 - (C) Liver cirrhosis
 - (D) Chronic inflammation
 - (E) Congestive heart failure
 - (F) All of the above
27. 缺，推測為(Which of the following is risk factor for thromboembolism?)
- (A) Atrial dilatation with fibrillation
 - (B) Aortic aneurysm
 - (C) Presence of anti-heparin/PF-4 IgG
 - (D) Factor V Leiden mutation
 - (E) All of the above
28. 缺，推測為(Which of the followings is commonly associated with septic shock?)
- (A) It can be with hyperinflammatory and/or immunosuppressive state.
 - (B) It frequently leads to DIC.
 - (C) Stress-induced hormone causes increased gluconeogenesis.
 - (D) All of the above
29. 缺
- (A) Adrenal cortical cell lipid accumulation
30. Which of the followings is correct?
- (A) Septic shock is most frequently triggered by Gram (+) bacterial infection.
 - (B) The pathogenesis of amniotic fluid embolism is primarily due to vascular obstruction.
 - (C) Lines of Zahn are seen in arterial thrombi, not in venous thrombi.
 - (D) Arterial thrombi tend to contain more RBC, and are thus known as red thrombi.
 - (E) Red infarcts occur in solid organ with end-arterial circulation.

B07 第 1 次區段考

1. Which is not considered as anatomical pathology?
 - (A) Surgical pathology
 - (B) Forensic pathology
 - (C) Chemical pathology
 - (D) Cytopathology
2. What is the most commonly used method for detecting gene amplification in pathological specimens?
 - (A) immunohistochemistry
 - (B) Polymerase chain reaction
 - (C) FISH
 - (D) Southern hybridization
3. At autopsy, the time of hospital autopsy by law is:
 - (A) 1hr
 - (B) 3hr
 - (C) 4hr
 - (D) 6hr
4. The purpose of hospital autopsy is based on:
 - (A) Doctor's academic interest
 - (B) Understanding of diseases
 - (C) Laboratory drug test
 - (D) Clinical experiment
5. What does autopsy mean for?
 - (A) Necropsy
 - (B) Postmortem dissection
 - (C) Dissection of organs
 - (D) See for oneself
6. Hospital autopsy needs:
 - (A) Consent of family member
 - (B) Consent of hospital
 - (C) Consent of coroner
 - (D) Consent of doctor

7. Which of the following is not true about ferroptosis:
(A) Iron
(B) Increased ROS
(C) Non-lytic cell death
(D) Inflammation
8. Which of the following descriptions about autophagy is wrong?
(A) Cancer cells may increase autophagy.
(B) Some neurodegenerative diseases are caused by insufficient autophagy.
(C) Autophagy is the main mechanism for recycling misfolded proteins.
(D) Autophagosome is double membrane structure.
9. Which of the following organs is not a common site for metastatic calcification:
(A) Stomach
(B) Kidney
(C) Pulmonary artery
(D) Lung
10. Which kind of adaption is the main mechanism of barrettes esophagus?
(A) Hyperplasia
(B) Metaplasia
(C) Hypertrophy
(D) Atrophy
11. Which inflammation is related to TB
(A) Caseating necrosis
(B) Coagulative necrosis
(C) Liquefactive necrosis
(D) Fat necrosis
12. Which one is irreversible cell injury
(A) Membrane rupture
(B) Cell swelling
(C) Eosinophilia
(D) Myelin figure
13. Which of the following pigments are NOT brown or yellow in color?
(A) Carbon
(B) Lipofuscin
(C) Melanin
(D) Hemosiderin

14. Which of the following molecules is not mainly involved in extrinsic pathway of apoptosis?
- (A) FasL
 - (B) FADD
 - (C) Caspase 8
 - (D) Apoptosome
15. 白血球從血管進到組織所需之各項分子配對，何者正確？
- (A) Rolling：selectin 和 sialyl-Lewis x glycoprotein
 - (B) Adhesion：I-CAM 和 selectin
 - (C) Transmigration：CD31 和 CD31
 - (D) Chemotaxis：LPS 和 G protein-coupled receptor
 - (E) 以上皆是
16. AA 是發炎重要的因子（中間一坨廢話），請問下列何者並非促進發炎？
- (A) Prostaglandin
 - (B) Leukotriene
 - (C) Thromboxane A2
 - (D) Lipoxin
17. 非類固醇消炎藥（NSAIDs）如阿斯匹靈（aspirin）能抑制發炎是因為它抑制了哪一種酵素，使 arachidonic acid metabolites 的合成降低？Which enzyme playing a critical role in inflammation can be inhibited by NSAIDs?
- (A) Cyclooxygenase
 - (B) 5-lipoxygenase
 - (C) Phospholipases
 - (D) 12-lipoxygenase
18. Steroid 能抑制發炎是因為抑制發炎了哪個酶？
- (A) 5-lipoxygenase
 - (B) Cyclooxygenase
 - (C) 12-lipoxygenase
 - (D) Phospholipase
19. 哪些敘述符合 neutrophil extracellular traps（NETs）？
- (A) Neutrophil 的 antimicrobial mechanism 之一
 - (B) 成份為 chromatin 及 granule 中的 enzymes
 - (C) 與 autoimmune disease 和感染相關的 thrombotic disease（thrombotic microangiopathy）有關聯性
 - (D) 細菌若帶有 DNase 則對 NETs 較有抗性
 - (E) 以上皆是
20. 缺

21. 下列哪一項不是 acute inflammation 的型態？
 - (A) Suppurative inflammation
 - (B) Fibrinoid inflammation
 - (C) Serous inflammation
 - (D) Granulomatous inflammation
22. Granuloma 在顯微鏡下看不到哪個東西？
 - (A) Histiocyte
 - (B) Giant cell
 - (C) Lymphocyte
 - (D) May have caseating necrosis
 - (E) Angiogenesis
23. 下列何者是 chronic inflammation 的特色？
 - (A) Monocytes infiltration
 - (B) Granulomatous inflammation
 - (C) Fibrosis and scar tissue
 - (D) 以上皆是
24. Macrophages 在 inflammation 很重要，下列何者不正確？
 - (A) INF- γ 活化 M1 macrophage
 - (B) IL-13, IL-4 活化 M2 macrophage
 - (C) M1 macrophage 製造 TGF- β
 - (D) M1 macrophage 和發炎有關
 - (E) M2 macrophage 和修復有關
25. 下列何種微生物所引起的發炎反應與眾不同？
 - (A) Tuberculosis
 - (B) Streptococcus pneumonia
 - (C) Leprosy
 - (D) Syphilis
26. Angiogenesis 中，tip cells 和 stalk cells 的分化和下列哪個 signaling pathway 有關？
 - (A) EGFR
 - (B) Dll4/Notch signal
 - (C) Wnt/beta-catenin
 - (D) MAPK/ERK pathway

27. 以下哪種病症和顯微鏡觀察下的發炎細胞配對錯誤？
- (A) 急性發炎：neutrophil
 - (B) 慢性發炎：lymphocyte
 - (C) Chronic B hepatitis：eosinophil
 - (D) Sarcoidosis：histocyte
28. Repair 中過度的 collagen 和 connective tissue deposition，會造成下列哪些情況？
- (A) Hypertrophic scar
 - (B) Keloid
 - (C) Contraction
 - (D) 以上皆是
29. 缺
30. Which of the following is/are associated with hypercoagulability?
- (1) Prothrombin mutation
 - (2) Trousseau syndrome
 - (3) Congestive heart failure
 - (4) Anti-phospholipid syndrome
 - (5) Bernard Soulier syndrome
- (A) (1), (2), (3), (4)
 - (B) (1), (2)
 - (C) (1), (2), (3), (5)
 - (D) (2), (5)
31. Which of the following is usually not present in a patient with advanced liver cirrhosis?
- (A) Ascites
 - (B) Edema
 - (C) Congestive splenomegaly
 - (D) Increased portal pressure
 - (E) Chronic passive congestion/necrosis

32. Which of the following factors have anti-thrombic activity?

- (1) Protein C
- (2) Antithrombin III
- (3) Tissue factor p inhibitor
- (4) Heparin-like molecule
- (5) Platelet activating factor 4
- (A) (1), (2), (3), (5)
- (B) (2), (3), (4), (5)
- (C) (1), (2), (3), (4)
- (D) (1), (3), (4), (5)
- (E) (1), (2), (3), (4), (5)

33. Which of the following statements are right?

- (1) Septic shock usually manifests as hyperglycemia.
- (2) Cardiogenic shock with large myocardial infarction usually carries poor prognosis.
- (3) DIC usually presents as early bleeding tendency. (裡面有 early 這個字，但我不確定放在哪)
- (4) Endothelial injury is a common cause of arterial thrombosis.
- (5) Warm and pale skin is an early sign of severe septic shock.
- (A) (1), (2), (4)
- (B) (1), (2), (3), (4), (5)
- (C) (1), (2), (4), (5)
- (D) (2), (4), (5)
- (E) (1), (2), (3), (4)

34. What is the result of thromboembolism

- (1) Red infarct
- (2) Liquefactive necrosis
- (3) Coagulative necrosis
- (4) Caseous necrosis
- (5) Cassion disease
- (A) (1), (2), (3), (4)
- (B) (1), (2), (3)
- (C) (1), (2), (4), (5)
- (D) (1), (2), (3), (5)
- (E) (1), (2), (3), (4), (5)

35. Which of the following finding(s) is found in patients with right heart failure? (多選)

- (1) Increased central venous pressure
- (2) Pulmonary edema
- (3) Lower leg edema
- (4) Dyspnea
- (5) Nutmeg liver

B08 第 1 次區段考

1. The following are commonly practiced by anatomical pathologists except:
 - (A) Surgical pathology
 - (B) Forensic pathology
 - (C) Chemical pathology
 - (D) Cytopathology
2. Which of the following methods is commonly used to detect gene amplification?
 - (A) Immunohistochemistry
 - (B) Immunofluorescence
 - (C) Fluorescent in situ hybridization
 - (D) Electromicroscopy
3. What is the meaning of autopsy?
 - (A) Necropsy
 - (B) See for oneself
 - (C) Post-Mortem biopsy
 - (D) Dissection of the body
4. Which is the sensor in the apoptosis pathway?
 - (A) BAK
 - (B) BAX
 - (C) BCL-2
 - (D) BID
5. Which one can be recognized by Annexin V of Annexin apoptotic kit?
 - (A) Propidium iodide
 - (B) Phosphatidylserine
 - (C) DAMP
 - (D) DNA
6. Which of the following organs is not a common site for metastatic calcification
 - (A) Stomach
 - (B) Kidney
 - (C) Pulmonary artery
 - (D) Lung

7. Which of the following pigments are mostly likely to be found in bruise areas ?
- (A) Carbon
 - (B) Lipofuscin
 - (C) Melanin
 - (D) Hemosiderin
8. Which kind of adaption is the main mechanism of cardiac enlargement in athletes?
- (A) Hyperplasia
 - (B) Hypertrophy
 - (C) Atrophy
 - (D) Metaplasia
9. 哪些不是 PAMPs ?
- (A) bacterial LPS
 - (B) bacterial peptidoglycan
 - (C) dsRNA
 - (D) interferon gamma
10. 下列關於急性發炎的敘述何者錯誤？
- (A) 組織水腫時通常是高蛋白濃度的 exudate
 - (B) 在發炎組織發現 neutrophil 及血管增生即可稱為 pus
 - (C) 在急性發炎初期經常發現 neutrophil 浸潤
 - (D) 在 neutrophil 的 granules 中含有大量 protease，若釋放到細胞外可能傷害組織
11. 下列有關急性肺炎與慢性發炎的敘述，何者正確？
- (A) Serous inflammation 常見富含蛋白質的 exudate
 - (B) Fibrinous inflammation 有大量 fibrin 沉積，常見於體腔壁，如：心包膜
 - (C) Suppurative inflammation 的特徵為巨噬細胞包圍壞死部分
 - (D) 慢性發炎的特徵為單核球、淋巴球浸潤與水腫
12. 關於發炎和巨噬細胞，下列何者正確？
- (A) Kupffer cell 是肝臟中的巨噬細胞，由骨髓中的 monocyte 發育而成
 - (B) 巨噬細胞因不同刺激發育成不同型態。其中 M1 促進發炎，M2 抑制發炎
 - (C) TH17 分泌 TGF-beta
 - (D) foreign body inflammation 中可以觀察到 granulomatous inflammation、histocyte infiltration、caseous necrosis

13. 關於傷口修復，以下何者錯誤？
- (A) 傷口修復時，引起血管新生的介質是 VEGF
 - (B) 增生期大量的微血管、纖維母細胞與疏鬆的間質，此即為肉芽組織(granulation tissue)
 - (C) 組織纖維化的介質是 TGF- β
 - (D) 傷口修復後，疤痕組織增生超過傷口範圍且不能自行恢復，此即為增生性疤痕(hypertrophic scar)
14. 音樂家蕭邦長年為肺病所苦，1849 年於法國病逝，死後根據其遺願，將其心臟放在酒精中，運回波蘭，他的死因一直未有定論。2017 年波蘭科學家將其心臟取出，推測他是因感染結核病 tuberculosis 造成的心包膜炎而死，請問對蕭邦的心臟做切片，會有什麼病理特徵？
- (A) 肉芽組織(granulation tissue)
 - (B) 肉芽腫發炎(granulation inflammation)
 - (C) 增生性疤痕(hypertrophic scar)
 - (D) 蟹足腫(keloid)
15. Which of the following factor have anti-thrombotic activity?
- (1) Protein C
 - (2) Thrombomodulin
 - (3) Tissue factor pathway inhibitor
 - (4) Heparin-like molecules
 - (5) Plasminogen activator inhibitor (PAI)
- (A) (1), (2), (3), (4)
 - (B) (2), (3), (4), (5)
 - (C) (1), (2), (3), (5)
 - (D) (1), (3), (4), (5)
 - (E) (1), (2), (3), (4), (5)
16. Which of the followings are results/complications of thrombosis/embolism?
- (1) white infarct
 - (2) liquefactive necrosis
 - (3) coagulative necrosis
 - (4) caseous necrosis
 - (5) Caisson disease
- (A) 1234
 - (B) 123
 - (C) 1245
 - (D) 1235
 - (E) 12345

17. Which of the following statements are right?
- (1) Patients with liver cirrhosis may manifest edema due to hyperalbuminemia.
 - (2) Acute tubular necrosis is a common result of shock.
 - (3) Disseminated intravascular coagulation initially presents as thrombosis tendency.
 - (4) Arterial thrombosis is usually caused by endothelial injury.
 - (5) Increase GLUT-4 may mediate hyperglycemia in septic shock.
- (A) 2, 3, 4
(B) 1, 2, 3, 4, 5
(C) 2, 4
(D) 2, 4, 5
(E) 2, 3, 4, 5
18. Which will be seen in patients of pure left heart failure?
- (1) increased central venous pressure
 - (2) brown induration of lung
 - (3) nutmeg liver
 - (4) pulmonary edema
 - (5) lower leg edema
- (A) 123
(B) 35
(C) 125
(D) 135
(E) 24
19. Which one is not related to anaphylaxis?
- (A) Hypotension due to vascular dilation
 - (B) Dyspnea due to laryngeal spasm
 - (C) Type I hypersensitivity
 - (D) Hypertension due to tachycardia
20. A 28-year-old young female with severe abdominal pain, anemia and died in the emergency room. Which autopsy finding is most likely? (非本次範圍)
- (A) Acute appendicitis
 - (B) Perforated peptic ulcer
 - (C) Uterine myoma
 - (D) Ectopic pregnancy

B09 第 1 次區段考

1. Which of the job is not a part of the daily work of anatomical pathologist?
 - (A) Cytology
 - (B) Frozen section
 - (C) Chemical pathology
 - (D) Molecular pathology
2. Which of the following methods is commonly used to detect protein expression?
 - (A) Immunohistochemistry
 - (B) Mass spectrometry (MS)
 - (C) In situ hybridization
 - (D) Electron microscopy (EM)
3. 台大醫院進行病理解剖的目的不包括下列何者？
 - (A) 釐清醫療糾紛的法律責任
 - (B) 住院醫師訓練
 - (C) 醫學系大五臨床病理討論會案例
 - (D) 學術研究
4. 下列關於病理科解剖的敘述何者正確？
 - (A) 不需經過家屬同意
 - (B) 病理科醫師當天即可發報告
 - (C) 取出的器官會用顯微鏡觀察
 - (D) 家屬要數年後才能領取遺體
5. Which kind of adaptation happens in liver after partial resection ?
 - (A) Metaplasia
 - (B) Atrophy
 - (C) Hypertrophy
 - (D) Hyperplasia
6. Which one is not a figure of reversible cell injury ?
 - (A) Myelin figure (EM)
 - (B) Surface blebbing (LM)
 - (C) Plasma membrane rupture (LM)
 - (D) Mitochondrial swelling (EM)

7. 下列何者正確？
- (A) ferroptosis 有 ROS 參與
 - (B) necrosome 在 necrosis 中扮演重要角色
 - (C) caspase 1 在 apoptosis 扮演重要角色
 - (D) pyroptosis 會分裂成多個 apoptotic body 並被 macrophage 吞噬
8. Which is the most common necrotic finding in tuberculosis infections?
- (A) Gangrenous necrosis
 - (B) Pancreatic fat necrosis
 - (C) Fibrinous necrosis
 - (D) Caseous necrosis
9. Which of the following is usually caused by misfolding protein response?
- (A) necrosis
 - (B) necroptosis
 - (C) apoptosis
 - (D) ferroptosis
10. 下列何者為 pro-apoptotic effector of intrinsic pathway
- (A) BCL2
 - (B) MCL2
 - (C) LC3
 - (D) BAX
11. Which of the following accumulation is seen in steatosis?
- (A) Ochronosis
 - (B) Lipofuscin
 - (C) Triglyceride
 - (D) Cholesterol
12. Which of the following molecules plays the most important role in anti-aging effect during starvation?
- (A) BCL-XL
 - (B) LC3
 - (C) Calcium
 - (D) Sirtuin
13. 關於組織胺對血管變化的描述，以下何者錯誤？
- (A) 組織胺作用於平滑肌，讓血管擴張
 - (B) 血管變化讓白血球能穿過血管壁，到達發炎組織
 - (C) 血管通透性增加，可能因為內皮細胞受損造成
 - (D) 血管通透性增加，蛋白和細胞穿過血管壁，稱為漏出液(transudate)

14. 在急性發炎中，關於白血球穿過血管內皮細胞的機制何者錯誤？
 - (A) Margination：白血球藉由 chemotaxis 附著到血管內皮細胞表面
 - (B) Rolling：白血球與內皮細胞藉由 selectin 與其 ligand 發生弱的交互作用
 - (C) Adhesion：白血球與內皮細胞藉由 integrin 與其 ligand 結合而強力黏著
 - (D) Diapedesis：白血球與內皮細胞藉由 CD31 而移行穿過血管壁
15. 急性發炎中 neutrophil 扮演重要角色，下列何者錯誤？
 - (A) neutrophil 只在 6~24 小時在組織中出現，macrophage 比較晚出現但存活較久
 - (B) 可以釋放染色質包住微生物，即 NET
 - (C) 主要藉由 iNOS 生成 NO 滅菌
 - (D) 可被 TNF、IL-1 活化
16. 以下敘述何者錯誤？
 - (A) 漿液性發炎 (serous inflammation) 好發於體腔，例如：心內膜、肋膜。
 - (B) 化膿性發炎 (suppurative inflammation) 可觀察到嗜中性球和壞死組織。
 - (C) 慢性發炎常可見淋巴球和巨噬細胞浸潤，且背景呈現纖維化。
 - (D) 嗜酸性球與寄生蟲感染和自體免疫有關
17. 下列何者錯誤？
 - (A) bradykinin 會造成 vasodilation，也與疼痛有關
 - (B) COX 產生的物質都會促進發炎，LOX 的都會抑制發炎
 - (C) C3a, C5a 會促進 neutrophil 的趨化作用
 - (D) 發炎時如果抽血發現 leukocytosis (核葉數 left shift)，是因為 TNF, IL-1 的刺激
18. 發炎反應的敘述何者錯誤？
 - (A) 急性發炎不一定會演變成慢性發炎
 - (B) 在 Th17 受損的病患，因為嗜中性球無法被活化，所以會導致 cold abscess 的產生。
 - (C) 在組織從發炎變成修復的過程中，巨噬細胞會從 M2 變成 M1 的型態。
 - (D) 體內也有內生性物質可以導致發炎反應
19. 組織修復時，會有細胞疏鬆、血管新生...的稱為什麼？
 - (A) 肉芽組織 granulation tissue
 - (B) 肉芽腫 granuloma
 - (C) 增生性疤痕 hypertrophic scar
 - (D) 蟹足腫 keloid

20. 肺臟組織切片看到 caseous necrosis、macrophage、lymphocyte、Langhans giant cell，可判斷為何種診斷？
- (A) Sarcoidosis
 - (B) 結核菌感染
 - (C) 膿瘍
 - (D) Foreign body
21. Which of the following are common symptoms and signs in a patient with right heart failure?
- (1) hepatomegaly
 - (2) pulmonary edema
 - (3) centrizonal congestion of liver
 - (4) increased hydrostatic pressure in pulmonary vein
 - (5) lower leg edema
- (A) 123
 - (B) 34
 - (C) 25
 - (D) 35
 - (E) 135
22. Which of the following factor have anti-thrombotic activity?
- (1) Protein C
 - (2) Thrombomodulin
 - (3) Tissue factor
 - (4) Heparin-like molecules
 - (5) Tissue plasminogen activator (tPA)
- (A) (1), (2), (4), (5)
 - (B) (1), (2), (3), (4)
 - (C) (1), (2), (3), (5)
 - (D) (1), (3), (4), (5)
 - (E) (1), (2), (3), (4), (5)

23. Which of the following is related to the pathogenesis of septic shock?

- (1) decreased tissue factor pathway inhibitor
- (2) DIC
- (3) peripheral pooling
- (4) myocardial depression
- (5) hyperglycemia
- (A) 1234
- (B) 1235
- (C) 1345
- (D) 2345
- (E) 12345

24. Which of the following statements are correct?

- (1) Patient with liver cirrhosis may manifest edema due to hypoalbuminemia
- (2) Acute tubular necrosis is usually a result of shock
- (3) DIC initially presents as hypercoagulability
- (4) Arterial thrombosis is usually caused by endothelial injury
- (5) Decreased GLUT-4 may cause hyperglycemia in shock
- (A) 1245
- (B) 1235
- (C) 1345
- (D) 2345
- (E) 12345

25. What is the result of thromboembolism?

- (1) Red infarct
- (2) Caseous necrosis
- (3) White infarct
- (4) Saddle embolism
- (5) Coagulative necrosis
- (A) 1234
- (B) 1345
- (C) 1235
- (D) 135
- (E) 12345

26. Which of the following is/are associated with hypercoagulability?

- (1) Migratory thrombophlebitis
- (2) Bernard-Soulier syndrome
- (3) Leiden mutation
- (4) von Willebrand disease
- (5) Anti-phospholipid syndrome

- (A) 1234
- (B) 1345
- (C) 1235
- (D) 135
- (E) 12345

B10 第 1 次區段考

1. (填充)
Patho-的意思?
-ology 的意思?
2. 下列何者為 brain autopsy 目的?
(A).確認可能之神經退化疾病的病理診斷。
(B).儲存腦組織以作為神經疾病研究之用。
(C).研究病患腦神經疾病的相關死因
(D).以上皆是
3. Which type of mechanism is included in the brain pathology of chronic ischemic dementia (vascular dementia)?
(A).Hyperplasia
(B).Hypertrophy
(C).Metaplasia
(D).Atrophy
4. Which type of necrotic pattern is typical of tuberculosis?
(A).coagulative necrosis
(B).liquefactive necrosis
(C).caseous necrosis
(D).gangrenous necrosis
5. Which of the following regarding cell death is TRUE?
(A).necrosome is the main mediated of necrosis
(B).the main caspase for ferroptosis is caspase 1
(C).pyroptosis won't cause inflammation
(D).necroptosis is triggered by death receptor
6. Which of the following morphological changes is NOT seen in cellular injury?
(A).surface bleb
(B).cytoplasmic eosinophilia
(C).myelin figure
(D).vacuolated cytoplasm
7. Which of the statement of ROS as a mechanism of cell injury is false?
(A).In hemostasis, no ROS can be found in cell.
(B).Mitochondrial injury will produce excess ROS.
(C).Excess ROS cause DNA damage.
(D).Catalase plays a role in the reduction of hydrogen peroxide.
8. Which of the following acts as anti-apoptotic molecule?
(A).BCL-2
(B).PUMA
(C).BAX
(D).BAK
9. 關於 autophagy 何者錯誤?
(A).Misfolded protein 最常用 autophagy 清理
(B).存在 Neurodegenerative disease 是 autophagy 過度活化的結果
(C).cancer 會使 autophagy 過度活化
(D).autophagosome 為單層膜

10. Which of the following accumulation is seen in steatosis?
(A).Ochronosis
(B).Triglyceride
(C).Lipofuscin
(D).Cholesterol
11. 下列關於發炎，何者錯誤？
(A).亡佚
(B).白血球透過血管新生進入發炎處
(C).急性發炎時嗜中性球會到發炎處吞噬病原
(D).急性發炎因為嗜中性球壽命短且 cytokines 半衰期短，因此病原體清除後自行結束發炎反應
12. 在急性發炎中，關於白血球穿過血管內皮細胞的機制何者錯誤？
(A).Margination：白血球藉由血流流速減緩附著到血管內皮細胞表面
(B).Rolling：白血球與內皮細胞藉由 selectin 與其 ligand 發生弱的交互作用
(C).Adhesion：白血球與內皮細胞藉由 integrin 與其 ligand 結合而強力黏著
(D).Chemotaxis：白血球的 GPCR 與 histamine 作用前往發炎區域
13. 急性發炎中 neutrophil 扮演重要角色，下列何者錯誤？
(A).neutrophil 只在 6~24 小時在組織中出現，macrophage 比較晚出現但存活較久
(B).可以釋放染色質包住微生物，即 NET
(C).主要藉由 iNOS 生成 NO 滅菌
(D).可被 TNF、IL-1 活化
14. 關於發炎反應的 Mediator 敘述下列何者錯誤？
(A).Arachidonic metabolites 有些抑制發炎，如 PGE2
(B).Bradykinin 與 substance P 和疼痛有關
(C).TNF 與 IL-1 和發燒有關
(D).補體、趨同反應、Phagocytosis 三者有關
15. 關於發炎反應何者錯誤？(選項亡佚)
(A).eosinophil 跟自體免疫與病毒相關
16. 手術數個月後，增生疤痕組織超出傷口範圍，病理切片下大量的 collagen，請問最可能是下列何者增生？
(A).granuloma
(B).granulation tissue
(C).keloid
(D).hypertrophic scar
17. 下列有關肉芽腫性發炎(granulomatous inflammation)的敘述何種錯誤？
(A).一種特殊型態的慢性發炎，通常是不易消滅的病菌所引發的反應
(B).組織特徵為巨噬細胞(Macrophage)及 B 淋巴球，有時會出現多核巨細胞(Multinucleated giant cell)
(C).有時會表現乾酪性壞死(Caseous necrosis)，有無壞死可以作為鑑別診斷的參考。
(D).難以分解的異物如手術縫線也可能造成肉芽性腫發炎。
18. 貝多芬於 1827 年去世，維也納醫師根據遺囑執行病理解剖，報告中提及腹腔中有 4 公升混濁積液。有學者推測是細菌性腹膜炎。若為真，該積液的描述為？
(A).很少蛋白質、很少白血球、應屬於漏出液(Transudate)
(B).大量蛋白質，很少白血球，應屬於漏出液(Transudate)
(C).大量蛋白質，大量白血球，應屬於滲出液(Exudate)
(D).很少蛋白質，大量白血球，應屬於滲出液(Exudate)

19. (多選) Which following description is correct about septic shock?
 - (A). microorganism pathogen-associated molecular-pattern
 - (B). increase thrombomodulin
 - (C). decrease t-PAi
 - (D). impaired GLUT-4
 - (E). insulin resistance
20. Which of the followings can be results/complications of thrombosis/embolus?
 1. red infarct
 2. septic infarction
 3. coagulative necrosis
 4. caseous necrosis
 5. bends
 - (A). 123
 - (B). 234
 - (C). 1235
 - (D). 1234
 - (E). 1245
21. Which followings are associated with hypercoagulability?
 1. Single nucleotide mutation of 3'-untranslated region of factor V gene.
 2. Disseminated cancer
 3. Atrial fibrillation
 4. Anti-phospholipid disease
 5. Bernard-soulier syndrome
 - (A). 2、3、4
 - (B). 2、3、4、5
 - (C). 1、2、4、5
 - (D). 1、2、3、4
 - (E). 1、3、4、5
22. Which of the following statements can be causes of edema?
 1. angiogenesis
 2. congestive heart failure
 3. hepatic cirrhosis
 4. filariasis
 5. venous thrombosis
 - (A). 亡佚
 - (B). 2345
 - (C). 亡佚
 - (D). 1234
 - (E). 12345
23. 下列哪些是抗凝血因子?
 1. protein S
 2. t-PA
 3. tissue factor pathway inhibitor
 4. heparin-like molecules
 5. platelet activating factor 4
 - (A). 1235
 - (B). 2345
 - (C). 1234
 - (D). 1345
 - (E). 12345

24. 下列何者者會發生在左心衰竭的病人？
1. Central venous pressure increase
 2. Pulmonary congestion
 3. Centrizonal congestion
 4. Pulmonary edema
 5. Lower leg edema
- (A).123
(B).35
(C).125
(D).135
(E).24